

Volume II · Chapter 7 — The Hugging Face Ecosystem · Theory

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-II-Chapter-7-context.md for the AI interaction log.

Prerequisites: Ch 6.

Learning outcomes — the student can: navigate the Hub; use transformers pipelines and AutoModel/AutoTokenizer; load and process data with datasets; run efficient inference; manage models and licenses locally.

Topics (theory) — 8 h:

#	Topic	h
7.1	The Hub: models, datasets, spaces; model cards & licenses	1
7.2	transformers: pipelines, AutoModel/AutoTokenizer, config	2
7.3	datasets: loading, map, streaming	2
7.4	Tokenizers, batching, padding/truncation	1
7.5	Inference: CPU/GPU, half precision, device management	1

#	Topic	h
7.6	The evaluate library & metrics basics	1

Datasets/tools: HF transformers, datasets, evaluate; PubMedBERT and similar.

Assessment: inference-pipeline deliverable (**50%**); dataset-processing lab (**30%**); quiz (**20%**). **Key decisions:** model choice/license; precision vs. memory; streaming vs. in-

memory datasets. **References:** plan §13 → *Transformers & the Hugging Face ecosystem*. **Hours:** Theory **8** + Lab **12** = **20**.